Challenges and Resolutions

AWS DevOps Technical Assignment

1. Bootstrapping the first ECS image

ECS requires a container image in ECR before a service task can start, while the ECR repository itself is provisioned by Terraform. The environment was therefore bootstrapped in two stages: create ECR first, push the initial image, then apply the complete infrastructure. Subsequent deployments are performed by GitHub Actions.

2. Private workload connectivity

Fargate tasks and the database run in private subnets. The application needs outbound access for image retrieval and logging without being publicly reachable. A NAT Gateway provides outbound connectivity; the Application Load Balancer remains the only public entry point.

3. GitHub Actions OIDC authentication

The workflow initially could not assume the AWS deployment role. Inspection of the issued OIDC claims showed that this GitHub organization uses a subject format containing immutable owner and repository IDs. The IAM trust policy was updated to match the exact subject for the production environment. This retained short-lived credentials and avoided storing AWS access keys in GitHub.

4. RDS Free Tier limits

RDS creation failed because the Free Tier account did not allow the original seven-day automated backup retention. The retention period was adjusted to one day, allowing a Single-AZ db.t3.micro PostgreSQL instance to be provisioned within the account constraints.

5. Database TLS connection

The application initially reported a self-signed certificate validation error when connecting to RDS. The connection configuration was adjusted for the assessment environment and the conflicting sslmode parameter was removed from the Secrets Manager connection string. The ECS service was then redeployed to read the updated secret. In production, the application should validate the AWS RDS CA bundle explicitly.

6. Terraform and CI/CD ownership

After GitHub Actions deployed a new ECS task definition, a Terraform run attempted to restore the original bootstrap task definition. The ECS service now ignores task-definition drift, making Terraform responsible for infrastructure and GitHub Actions responsible for application revisions.

Conclusion

The resulting environment provides private compute and database tiers, a public load balancer, centralized logs and dashboards, and an OIDC-based CI/CD path. Cost-bearing resources such as the NAT Gateway and RDS instance should be destroyed after the assessment demonstration.